

EXPERIMENT- 03

Python Code:

```
03.py > ...
1 keyword = input("Enter keyword: ").upper().replace("J", "I")
2 plaintext = input("Enter plaintext: ").upper().replace("J", "I").replace(" ", "")
3
4 matrix = []
5 used = set()
6 for char in keyword + "ABCDEFGHIKLMNOPQRSTUVWXYZ":
7     if char not in used:
8         matrix.append(char)
9         used.add(char)
10 matrix = [matrix[i:i+5] for i in range(0, 25, 5)]
11
12 def get_pos(c):
13     for i in range(5):
14         for j in range(5):
15             if matrix[i][j] == c:
16                 return i, j
17
18 prepared_text = ""
19 i = 0
20 while i < len(plaintext):
21     a = plaintext[i]
22     if i + 1 < len(plaintext):
23         b = plaintext[i+1]
24         if a == b:
25             prepared_text += a + "X"
26             i += 1
27         else:
28             prepared_text += a + b
29             i += 2
30     else:
31         prepared_text += a + "X"
32         i += 1
33
34 encrypted_text = ""
35 for i in range(0, len(prepared_text), 2):
36     r1, c1 = get_pos(prepared_text[i])
37     r2, c2 = get_pos(prepared_text[i+1])
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38
39     if r1 == r2:
40         encrypted_text += matrix[r1][(c1+1)%5] + matrix[r2][(c2+1)%5]
41     elif c1 == c2:
42         encrypted_text += matrix[(r1+1)%5][c1] + matrix[(r2+1)%5][c2]
43     else:
44         encrypted_text += matrix[r1][c2] + matrix[r2][c1]
45
46     print("Encrypted Text:", encrypted_text)
47
48     decrypted_text = ""
49     for i in range(0, len(encrypted_text), 2):
50         r1, c1 = get_pos(encrypted_text[i])
51         r2, c2 = get_pos(encrypted_text[i+1])
52
53         if r1 == r2:
54             decrypted_text += matrix[r1][(c1-1)%5] + matrix[r2][(c2-1)%5]
55         elif c1 == c2:
56             decrypted_text += matrix[(r1-1)%5][c1] + matrix[(r2-1)%5][c2]
57         else:
58             decrypted_text += matrix[r1][c2] + matrix[r2][c1]
59
60     print("Decrypted Text:" decrypted_text)

```

Output:

```

PS D:\College\Crypto\Experiments> python -u "d:\College\Crypto\Experiments\03.py"
Enter keyword: MONARCHY
Enter plaintext: INSTRUMENTS
Encrypted Text: GATLMZCLRQXA
Decrypted Text: INSTRUMENTSX
PS D:\College\Crypto\Experiments>

```